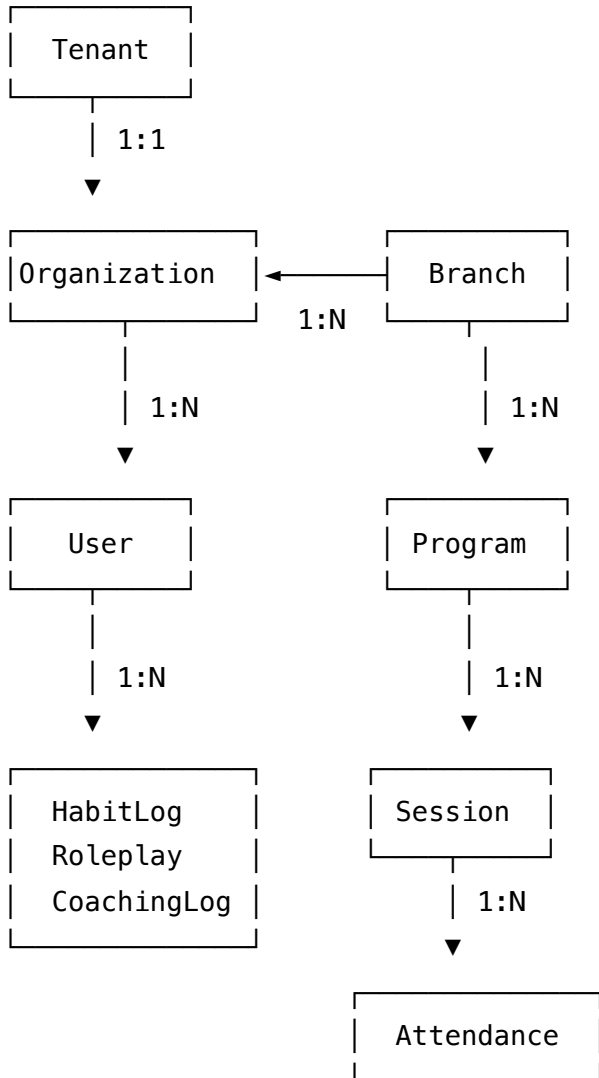


E2E Service System - Data Model & Schema

1. Entity Relationship Diagram



2. Core Entities

2.1 Tenant

```
CREATE TABLE tenants (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name VARCHAR(255) NOT NULL,  
  subdomain VARCHAR(100) UNIQUE NOT NULL,  
  plan VARCHAR(50) NOT NULL, -- Starter, Growth, Enterprise  
  status VARCHAR(50) DEFAULT 'Active', -- Active, Suspended, Cancelled  
  locale VARCHAR(10) DEFAULT 'en',  
  timezone VARCHAR(50) DEFAULT 'Africa/Accra',  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

2.2 Organization

```
CREATE TABLE organizations (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL REFERENCES tenants(id) ON DELETE CASCADE,  
  name VARCHAR(255) NOT NULL,  
  industry VARCHAR(100),  
  targets JSONB, -- {complaints: -20, csat: 2, adoption: 80}  
  baseline_metrics JSONB, -- {complaints: 50, csat: 3.5}  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_org_tenant ON organizations(tenant_id);
```

2.3 Branch

```
CREATE TABLE branches (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  org_id UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE,  
  name VARCHAR(255) NOT NULL,  
  region VARCHAR(100),  
  manager_id UUID,  
  staff_count INT DEFAULT 0,  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE INDEX idx_branch_tenant ON branches(tenant_id, org_id);
```

2.4 User

```
CREATE TYPE user_role AS ENUM (  
    'SuperAdmin',  
    'ClientAdmin',  
    'Manager',  
    'Coach',  
    'Staff',  
    'ExecutiveViewer'  
);
```

```
CREATE TABLE users (  
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
    tenant_id UUID NOT NULL,  
    org_id UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE,  
    branch_id UUID REFERENCES branches(id),  
    role user_role NOT NULL,  
    email VARCHAR(255) NOT NULL,  
    password_hash VARCHAR(255) NOT NULL,  
    first_name VARCHAR(100),  
    last_name VARCHAR(100),  
    phone VARCHAR(20),  
    status VARCHAR(50) DEFAULT 'Active',  
    last_login_at TIMESTAMP,  
    created_at TIMESTAMP DEFAULT NOW(),  
    updated_at TIMESTAMP DEFAULT NOW(),  
    UNIQUE(tenant_id, email)  
);
```

```
CREATE INDEX idx_user_tenant ON users(tenant_id, email);
```

```
CREATE INDEX idx_user_branch ON users(branch_id);
```

3. Program & Session Entities

3.1 Program

```
CREATE TABLE programs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  org_id UUID NOT NULL REFERENCES organizations(id) ON DELETE CASCADE,  
  template_id UUID, -- Reference to global template  
  name VARCHAR(255) NOT NULL,  
  start_date DATE NOT NULL,  
  end_date DATE NOT NULL,  
  targets JSONB, -- {complaints: -20, csat: 2}  
  status VARCHAR(50) DEFAULT 'Active',  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE INDEX idx_program_tenant ON programs(tenant_id, org_id);
```

3.2 Session

```
CREATE TYPE session_type AS ENUM ('Huddle', 'Roleplay', 'Clinic', 'Workshop');
```

```
CREATE TABLE sessions (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  program_id UUID REFERENCES programs(id) ON DELETE CASCADE,  
  branch_id UUID REFERENCES branches(id),  
  type session_type NOT NULL,  
  title VARCHAR(255) NOT NULL,  
  agenda TEXT,  
  scheduled_at TIMESTAMP NOT NULL,  
  duration_min INT DEFAULT 15,  
  facilitator_id UUID REFERENCES users(id),  
  materials JSONB, -- [{name: 'slides.pdf', url: 'https://...'}]  
  status VARCHAR(50) DEFAULT 'Scheduled',  
  created_at TIMESTAMP DEFAULT NOW(),  
  updated_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_session_tenant ON sessions(tenant_id, branch_id);  
CREATE INDEX idx_session_date ON sessions(scheduled_at);
```

3.3 Attendance

```
CREATE TABLE attendance (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  session_id UUID NOT NULL REFERENCES sessions(id) ON DELETE CASCADE,  
  user_id UUID NOT NULL REFERENCES users(id),  
  attended BOOLEAN DEFAULT FALSE,  
  notes TEXT,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_attendance_tenant ON attendance(tenant_id, session_id);
```

4. Roleplay & Habits

4.1 Roleplay

```
CREATE TABLE roleplays (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  session_id UUID REFERENCES sessions(id),  
  user_id UUID NOT NULL REFERENCES users(id),  
  scenario_id UUID, -- Reference to scenario library  
  score INT CHECK (score >= 1 AND score <= 5),  
  coach_notes TEXT,  
  video_url TEXT,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_roleplay_tenant ON roleplays(tenant_id, user_id);
```

4.2 MicroHabit

```
CREATE TABLE micro_habits (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  org_id UUID NOT NULL REFERENCES organizations(id),  
  title VARCHAR(255) NOT NULL,  
  description TEXT,  
  frequency VARCHAR(50) DEFAULT 'Daily',  
  points INT DEFAULT 10,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_habit_tenant ON micro_habits(tenant_id, org_id);
```

4.3 HabitLog

```
CREATE TABLE habit_logs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  habit_id UUID NOT NULL REFERENCES micro_habits(id),  
  user_id UUID NOT NULL REFERENCES users(id),  
  date DATE NOT NULL,  
  completed BOOLEAN DEFAULT FALSE,  
  proof_url TEXT,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_habitlog_tenant ON habit_logs(tenant_id, user_id, date);
```

5. Coaching & Recognition

5.1 CoachingLog

```
CREATE TABLE coaching_logs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  manager_id UUID NOT NULL REFERENCES users(id),  
  staff_id UUID NOT NULL REFERENCES users(id),  
  date DATE NOT NULL,  
  topic VARCHAR(255),  
  notes TEXT,  
  evidence_url TEXT,  
  approved BOOLEAN DEFAULT FALSE,  
  approved_by UUID REFERENCES users(id),  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_coaching_tenant ON coaching_logs(tenant_id, manager_id);
```


5.2 Recognition

```
CREATE TABLE recognitions (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  from_user_id UUID NOT NULL REFERENCES users(id),  
  to_user_id UUID NOT NULL REFERENCES users(id),  
  tag VARCHAR(100), -- 'Empathy', 'Teamwork', etc.  
  note TEXT,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_recognition_tenant ON recognitions(tenant_id, to_user_id);
```

6. Metrics & Alerts

6.1 Metric

```
CREATE TYPE metric_type AS ENUM ('CSAT', 'Complaints', 'RepeatCustomer', 'TimeToResolve
```

```
CREATE TABLE metrics (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  branch_id UUID REFERENCES branches(id),  
  type metric_type NOT NULL,  
  value DECIMAL(10,2) NOT NULL,  
  date DATE NOT NULL,  
  source VARCHAR(100), -- 'Manual', 'Zendesk', 'Freshdesk'  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

```
CREATE INDEX idx_metric_tenant ON metrics(tenant_id, branch_id, type, date);
```

6.2 KPAlert

```
CREATE TABLE kpi_alerts (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  org_id UUID REFERENCES organizations(id),  
  metric_type metric_type NOT NULL,  
  threshold DECIMAL(10,2),  
  status VARCHAR(50) DEFAULT 'Active',  
  triggered_at TIMESTAMP,  
  resolved_at TIMESTAMP,  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE INDEX idx_alert_tenant ON kpi_alerts(tenant_id, status);
```

7. Templates & Billing

7.1 Template (Global)

```
CREATE TABLE templates (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  name VARCHAR(255) NOT NULL,  
  type VARCHAR(50), -- 'Pilot', 'Huddle', 'Coaching'  
  content JSONB,  
  is_global BOOLEAN DEFAULT TRUE,  
  created_by UUID, -- Super Admin  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

7.2 Subscription

```
CREATE TABLE subscriptions (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL REFERENCES tenants(id),  
  plan VARCHAR(50) NOT NULL,  
  status VARCHAR(50) DEFAULT 'Active',  
  billing_cycle VARCHAR(50) DEFAULT 'Monthly',  
  amount DECIMAL(10,2),  
  stripe_subscription_id VARCHAR(255),  
  current_period_start DATE,  
  current_period_end DATE,  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE INDEX idx_subscription_tenant ON subscriptions(tenant_id);
```

7.3 Usage

```
CREATE TABLE usage_logs (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  tenant_id UUID NOT NULL,  
  metric VARCHAR(100), -- 'active_users', 'sms_credits', 'storage_gb'  
  value INT,  
  date DATE NOT NULL,  
  created_at TIMESTAMP DEFAULT NOW()  
);  
  
CREATE INDEX idx_usage_tenant ON usage_logs(tenant_id, date);
```

8. Prisma Schema

```
generator client {
  provider = "prisma-client-js"
}

datasource db {
  provider = "postgresql"
  url      = env("DATABASE_URL")
}

model Tenant {
  id          String   @id @default(uuid())
  name        String
  subdomain   String   @unique
  plan        String
  status      String   @default("Active")
  locale      String   @default("en")
  timezone    String   @default("Africa/Accra")
  createdAt   DateTime @default(now()) @map("created_at")
  updatedAt   DateTime @updatedAt @map("updated_at")

  organizations Organization[]
  users           User[]
  branches        Branch[]

  @@map("tenants")
}

model Organization {
  id          String   @id @default(uuid())
  tenantId    String   @map("tenant_id")
  name        String
  industry    String?
  targets     Json?
  baselineMetrics Json? @map("baseline_metrics")
  createdAt   DateTime @default(now()) @map("created_at")
  updatedAt   DateTime @updatedAt @map("updated_at")

  tenant      Tenant    @relation(fields: [tenantId], references: [id], onDelete: Cascade)
  branches    Branch[]
  users       User[]
}
```

```

programs Program[]

@@index([tenantId])
@@map("organizations")
}

model Branch {
    id          String    @id @default(uuid())
    tenantId    String    @map("tenant_id")
    orgId       String    @map("org_id")
    name        String
    region      String?
    managerId   String?   @map("manager_id")
    staffCount  Int       @default(0) @map("staff_count")
    createdAt   DateTime  @default(now()) @map("created_at")
    updatedAt   DateTime  @updatedAt @map("updated_at")

    tenant      Tenant     @relation(fields: [tenantId], references: [id])
    organization Organization @relation(fields: [orgId], references: [id], onDelete: Cascade)
    users       User[]
    sessions    Session[]
    metrics     Metric[]

    @@index([tenantId, orgId])
    @@map("branches")
}

enum UserRole {
    SuperAdmin
    ClientAdmin
    Manager
    Coach
    Staff
    ExecutiveViewer
}

model User {
    id          String    @id @default(uuid())
    tenantId    String    @map("tenant_id")
    orgId       String    @map("org_id")
    branchId    String?   @map("branch_id")
    role        UserRole
    email       String

```

```

passwordHash String    @map("password_hash")
firstName     String?   @map("first_name")
lastName      String?   @map("last_name")
phone         String?
status        String    @default("Active")
lastLoginAt   DateTime? @map("last_login_at")
createdAt     DateTime  @default(now()) @map("created_at")
updatedAt     DateTime  @updatedAt @map("updated_at")

tenant        Tenant      @relation(fields: [tenantId], references: [id])
organization  Organization @relation(fields: [orgId], references: [id], onDelete: Ca
branch        Branch?     @relation(fields: [branchId], references: [id])
habitLogs     HabitLog[]
roleplays     Roleplay[]
coachingLogs  CoachingLog[] @relation("Coach")

@@unique([tenantId, email])
@@index([tenantId, email])
@@index([branchId])
@@map("users")
}

model Program {
  id          String    @id @default(uuid())
  tenantId    String    @map("tenant_id")
  orgId       String    @map("org_id")
  templateId  String?   @map("template_id")
  name        String
  startDate   DateTime  @map("start_date")
  endDate     DateTime  @map("end_date")
  targets     Json?
  status      String    @default("Active")
  createdAt   DateTime  @default(now()) @map("created_at")
  updatedAt   DateTime  @updatedAt @map("updated_at")

  organization Organization @relation(fields: [orgId], references: [id], onDelete: Casc
  sessions     Session[]

  @@index([tenantId, orgId])
  @@map("programs")
}

// Continue with other models...

```

9. Row-Level Security Policies

```
-- Enable RLS on all tenant-scoped tables
ALTER TABLE organizations ENABLE ROW LEVEL SECURITY;
ALTER TABLE branches ENABLE ROW LEVEL SECURITY;
ALTER TABLE users ENABLE ROW LEVEL SECURITY;

-- Create policy for tenant isolation
CREATE POLICY tenant_isolation_policy ON organizations
    USING (tenant_id = current_setting('app.current_tenant')::UUID);

CREATE POLICY tenant_isolation_policy ON branches
    USING (tenant_id = current_setting('app.current_tenant')::UUID);

CREATE POLICY tenant_isolation_policy ON users
    USING (tenant_id = current_setting('app.current_tenant')::UUID);
```

10. Sample Data

```
-- Insert sample tenant
INSERT INTO tenants (id, name, subdomain, plan) VALUES
('550e8400-e29b-41d4-a716-446655440000', 'Acme Corp', 'acme', 'Growth');

-- Insert organization
INSERT INTO organizations (id, tenant_id, name, industry, targets) VALUES
('660e8400-e29b-41d4-a716-446655440000', '550e8400-e29b-41d4-a716-446655440000',
'Acme Corp', 'Banking', '{"complaints": -20, "csat": 2}');

-- Insert branches
INSERT INTO branches (tenant_id, org_id, name, region) VALUES
('550e8400-e29b-41d4-a716-446655440000', '660e8400-e29b-41d4-a716-446655440000',
'Accra Main', 'Greater Accra'),
('550e8400-e29b-41d4-a716-446655440000', '660e8400-e29b-41d4-a716-446655440000',
'Kumasi Branch', 'Ashanti');
```